

Mohamed Mahmoud

Civil Engineering Graduate | Geotechnical & Site Engineering

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PROFILE

Civil Engineering graduate (S.T., Hasanuddin University, 2026) with hands-on experience in geotechnical testing, structural site inspection, and technical reporting. Conducted independent research on bacterial soil stabilisation using *Bacillus Tropicus*, covering a full geotechnical laboratory program including CBR testing, soil characterization, and data analysis. Completed a structural construction internship with daily site inspection, quality control, and contractor coordination on a multi-storey building project. Proficient in AutoCAD for professional technical drawings; experienced in digital tools and self-directed project development. Multilingual: Arabic (native), English (fluent), Indonesian (working), French (elementary).

EDUCATION

Bachelor of Civil Engineering (S.T.)

Aug 2021 – Feb 2026

Hasanuddin University, Makassar, Indonesia

GPA: 3.33 / 4.00 | Sangat Memuaskan (Very Satisfactory) | International Scholarship Recipient

Accreditation: University — Excellent (BAN-PT); Study Programme — Excellent (BAN-PT/ABET)

Key Modules: Geotechnical Engineering, Soil Mechanics, Foundation Engineering, Structural Analysis, Concrete Technology, Construction Management, Road Engineering, Project Management, Surveying

Dissertation: "The effect of curing time on the California bearing ratio (CBR) value" — independent geotechnical research covering soil characterization, laboratory testing, CBR analysis, and technical reporting.

EXPERIENCE

Civil Engineering Intern — Structural Construction

Mar 2025 – Jun 2025

PT. Ikram Tiga Berlian | Justice Tower Project, UNHAS Campus, Makassar

- Carried out daily site inspections of columns, beams, and slab works, verifying workmanship against approved drawings and design specifications.
- Performed slump tests and compressive strength sampling; maintained quality control records of concrete batching and material testing results.
- Supervised reinforcement placement, formwork installation, and concrete curing operations, ensuring compliance with construction standards and project schedules.
- Liaised with site engineers and foremen on construction progress, identifying non-conformances and supporting corrective actions.
- Prepared site inspection and quality control reports, maintaining complete project documentation throughout the construction program.

Undergraduate Researcher — Geotechnical Engineering

Aug 2024 – Feb 2026

Hasanuddin University, Makassar

- Designed and executed a comprehensive geotechnical testing program covering soil characterization and stabilization: specific gravity, sieve analysis, Atterberg limits, moisture content, Standard Proctor compaction, and soaked/unsoaked CBR.
- Applied *Bacillus Tropicus* bacterial treatment to SP-SC classified soil samples across multiple curing periods; conducted comparative CBR analysis between treated and untreated specimens.
- Produced a technical report with full methodology, laboratory data tables, analysis, and evidence-based conclusions. Findings are currently being prepared for journal submission.

PROJECTS & PORTFOLIO

Personal Website2025 – Present

mohamedmohamedm2003-cyber.github.io

- Designed and deployed a personal portfolio website using GitHub Pages, showcasing engineering work, research, and professional profile.
- Reflects self-driven approach to professional visibility and digital skills development.

GeoCore — Geotechnical Engineering Platform2025 – Present

Live demo: <https://uigen-30158.web.app/dashboard> (development build)

- Independently developing GeoCore, an advanced web-based geotechnical engineering platform for soil technicians and engineers, featuring soil test records, AI-assisted analysis, and a live engineering news feed. Built with React, TypeScript, and Firebase; currently in advanced development stage.

ADDITIONAL EXPERIENCE

Community Service Program2025

Indonesia

- Contributed to planning and design of small-scale construction works (housing and shelter units).
- Facilitated technical education sessions for community members, applying communication and field coordination skills.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AI Fluency for Students — Anthropic, University College Cork & National Forum (2026)
- Claude Code in Action — Anthropic (March 2026)
- Tech Success Roadmap Workshop: Between Global Vision and Local Needs — Sudanese Students Association, Purwokerto, Indonesia (Feb 2026)

TECHNICAL SKILLS

Geotechnical:	CBR testing (soaked & unsoaked), soil stabilization, geotechnical laboratory analysis, Standard Proctor compaction, sieve analysis, Atterberg limits, specific gravity, moisture content
Site Engineering:	Construction inspection, quality control, concrete testing (slump & compressive strength), reinforcement and formwork supervision
CAD & Drawing:	AutoCAD — floor plans, layouts, technical drawings (professional freelance level); AutoCAD Civil 3D (learning)
Software:	Microsoft Office (Word, Excel, PowerPoint); Microsoft Project (studied); GitHub Pages
Reporting:	Technical report writing, site inspection records, quality control documentation, progress reporting
Digital:	Web development (basic); App development (in progress); AI tools proficiency

LANGUAGES

Arabic:	Native
English:	Fluent
Indonesian:	Working proficiency
French:	Elementary (A1–A2, ongoing)

AWARDS

- International Scholarship — Hasanuddin University (full undergraduate scholarship, 2021–2026)
- Graduation Predicate: Sangat Memuaskan (Very Satisfactory) — GPA 3.33/4.00